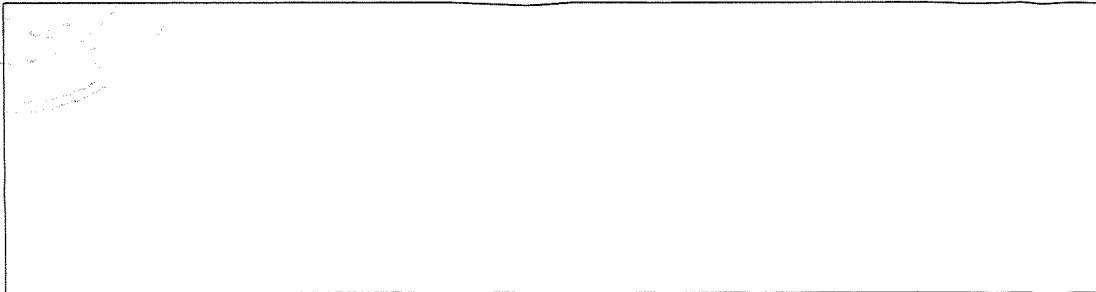




1	
2	
3	
4	
Total	

Question 1.

- (a) Write down the outcome space of the random experiment consists of tossing a fair coin, a double headed coin and a double-tailed coin simultaneously.



(5 Marks)

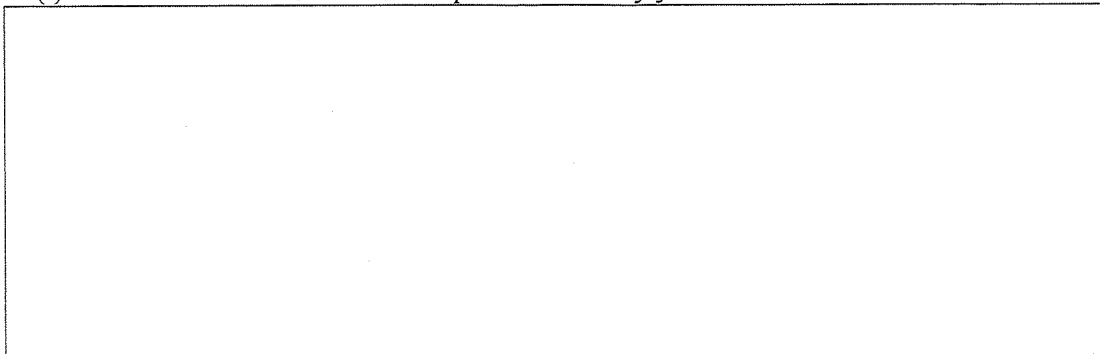
- (b) Toss a fair coin twice and observe the sequence of Heads and Tails. The following events were defined.

H_1 = Heads on first toss

H_2 = Heads on second toss

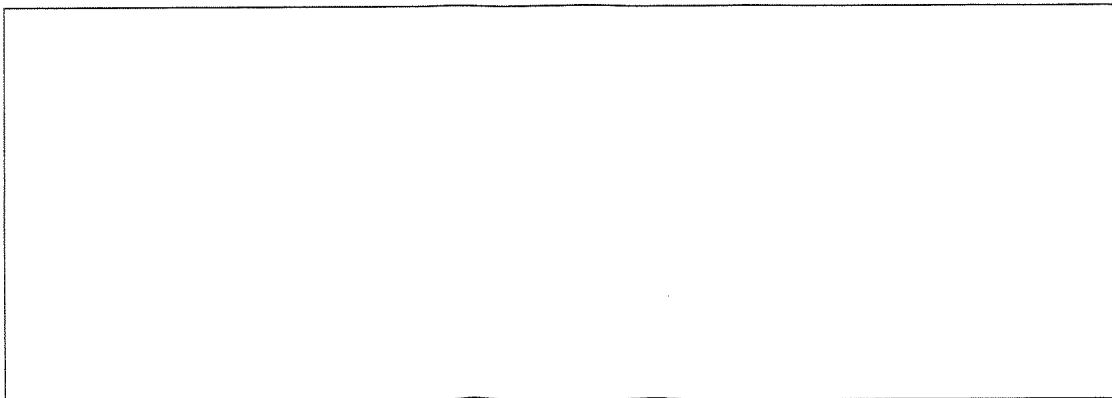
E = Two heads or two tails

- (i) Are the events H_1 and H_2 independent? Justify your answer



(5 Marks)

- (ii) Are the events H_1 , H_2 and E mutually independent? Justify your answer.



(5 Marks)

(c) Three production lines contribute to the total pool of a company's product. Line 1 provides 20% to the pool and 15% of its products are defective. Line 2 provides 50% to the pool and 5% of its products are defective. Line 3 contribute 30% to the pool and 6% of its products are defective.

(i) Calculate the percentage of items produced are defective.

(5 Marks)

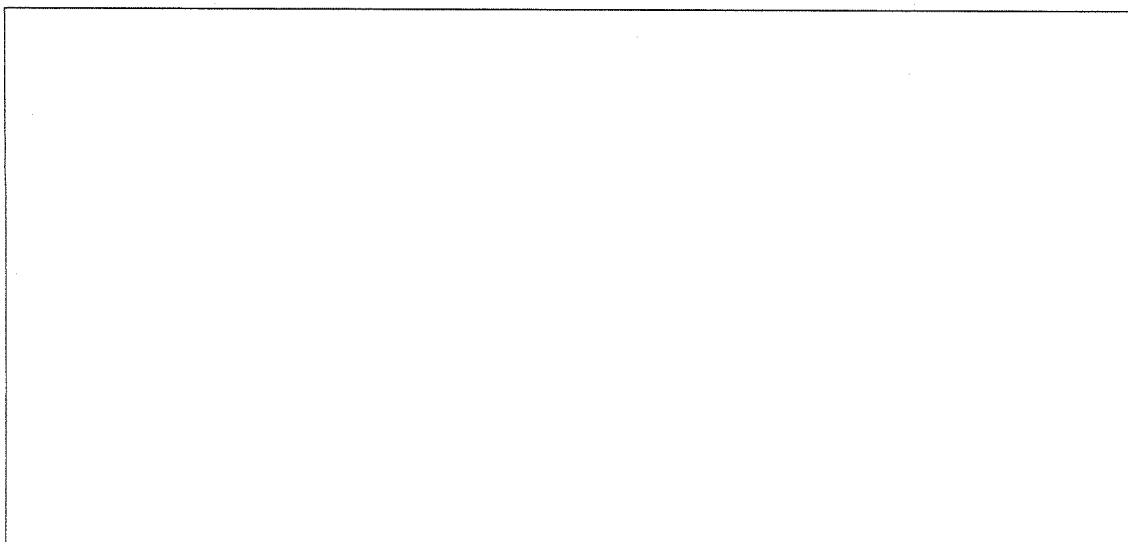
(ii) Suppose that an item is randomly picked from the pool and found to be defective. Calculate the probability that it came from Line 1.

(5 Marks)

Question 2.

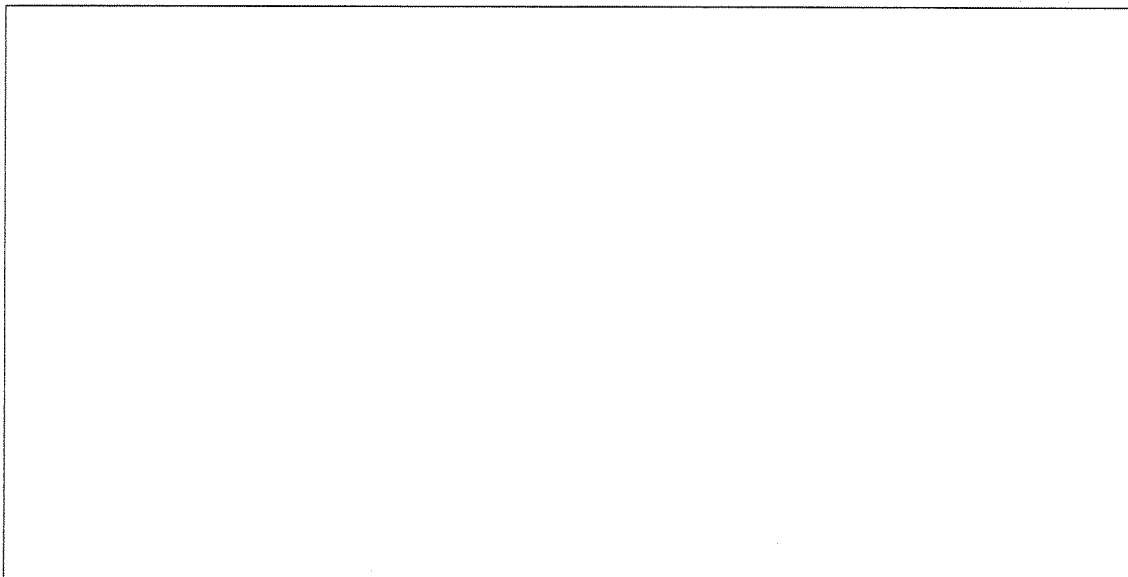
- (a) The probability that a bird watcher sees a kingfisher on any particular day is 0.3. Let X be the number of days in a week she sees a kingfisher.

(i) What is the distribution of X ? Justify your answer.



(4 Marks)

(ii) Calculate the probability that, in a week she sees a kingfisher exactly on 2 days.



(4 Marks)

- (b) A telephone operator receives calls independently of one another. Usually she receives in average 2.5 calls per hour on a working day. Calculate the probability that the operator receives more than 2 calls on a working day between 9.00am and 11.00am.

(6 Marks)

- (c) Among the 120 applicants for a job, only 80 are actually qualified. If five of the applicants are randomly selected for an in-depth interview, calculate the probability that only two will be actually qualified for the job. [Simplification is not expected]

(5 Marks)

- (d) A test consists of 10 multiple choice questions and there are four answers to each question of which only one is correct. If a student answers each question by rolling a uniform tetrahedral die (a die has only four sides), calculate the probability that he will get exactly half of the questions correct?

(6 Marks)

Question 3.

- (a) Toss an unbiased coin until either two heads or two tails (no need to be consecutively) have been occurred. You have to pay Rs.6.00 to start and receive Rs.5.00 for every head observed. Calculate the expected amount you get.

(8 Marks)

- (b) Let X be the number of heads obtained in four tosses of a fair coin, determine the probability distribution of $Y = \frac{1}{1+X}$.

(8 Marks)

- (c) The probability distribution of a random variable X is given in the table below.

X	0	2	4	6
$P(X=x)$	$3/8$	$5/16$	$4p$	p

- (i) Determine the value of p .

(5 Marks)

- (ii) Calculate the probability that X takes a value greater than 2?

(4 Marks)

Question 4.

- (a) A soft-drink vending machine is set so that the amount of drink dispensed is a normally distributed random variable with a mean of 200ml and standard deviation of 15ml. Calculate the probability that the average amount dispensed in a random sample of size 36 is at least 204ml.

(6 Marks)

- (b) It is given that the heights of adult females are normally distributed with mean 162cm and standard deviation 8cm. It is also given that the heights of adult males are normally distributed with mean 175cm with standard deviation 7cm.
- (i) Calculate the probability that a randomly picked adult female has a height less than 150cm.

(5 marks)

- (ii) Calculate the probability that a randomly picked adult male is taller than a randomly picked adult female?

(7 Marks)

- (c) Let $\bar{X}$ be the mean of a random sample with size 20 taken from a $N(100,160)$ distribution and $\bar{Y}$ be the mean of a random sample of size 40 taken from a $N(103,120)$ distribution.

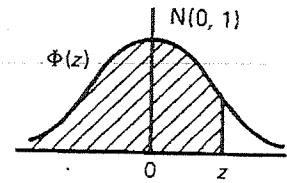
- (i) Write down the sampling distributions of $\bar{X}$ and $\bar{Y}$.

(4 Marks)

- (ii) If X and Y are independent, write down the sampling distribution of $\bar{Y} - \bar{X}$.

(3 Marks)

THE DISTRIBUTION FUNCTION $\Phi(z)$ OF
THE NORMAL DISTRIBUTION $N(0, 1)$



z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
											ADD								
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359	4	8	12	16	20	24	28	32	36
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753	4	8	12	16	20	24	28	32	36
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141	4	8	12	15	19	23	27	31	35
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517	4	7	11	15	19	22	26	30	34
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879	4	7	11	14	18	22	25	29	32
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224	3	7	10	14	17	20	24	27	31
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549	3	7	10	13	16	19	23	26	29
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852	3	6	9	12	15	18	21	24	27
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133	3	5	8	11	14	16	19	22	25
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389	3	5	8	10	13	15	18	20	23
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621	2	5	7	9	12	14	16	19	21
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830	2	4	6	8	10	12	14	16	18
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015	2	4	6	7	9	11	13	15	17
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177	2	3	5	6	8	10	11	13	14
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319	1	3	4	6	7	8	10	11	13
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441	1	2	4	5	6	7	8	10	11
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545	1	2	3	4	5	6	7	8	9
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633	1	2	3	4	4	5	6	7	8
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706	1	1	2	3	4	4	5	6	6
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767	1	1	2	2	3	4	4	5	5
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817	0	1	1	2	2	3	3	4	4
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857	0	1	1	2	2	2	3	3	4
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890	0	1	1	1	2	2	2	3	3
2.3	.9893	.9896	.9898								0	1	1	1	1	2	2	2	2
			.9901	.99036	.99061	.99086					3	5	8	10	13	15	18	20	23
2.4	.99180	.99202	.99224	.99245	.99266			.99111	.99134	.99158	2	5	7	9	12	14	16	18	21
					.99286	.99305	.99324	.99343	.99361		2	4	6	7	9	11	13	15	17
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520	2	3	5	6	8	9	11	12	14
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643	1	2	3	5	6	7	8	9	10
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736	1	2	3	4	5	6	7	8	9
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807	1	1	2	3	4	4	5	6	6
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861	0	1	1	2	2	3	3	4	4
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900	0	1	1	2	2	2	3	3	4
3.1	.99932	.99965	.99996								3	6	9	13	16	19	22	25	28
			.999126	.999155	.999184	.999211					3	6	8	11	14	17	20	22	25
3.2	.999313	.999336	.999359	.999381	.999402			.999238	.999264	.999289	2	5	7	10	12	15	17	20	22
					.999423	.999443	.999462	.999481	.999499		2	4	7	9	11	13	15	18	20
3.3	.999517	.999534	.999550	.999566	.999581						2	4	6	8	9	11	13	15	17
					.999596	.999610	.999624	.999638	.999651		2	3	5	6	8	10	11	13	14
3.4	.999663	.999675	.999687	.999698	.999709	.999720	.999730	.999740	.999749	.999758	1	3	4	5	7	8	9	10	12
											1	2	3	4	5	6	7	8	9
3.5	.999767	.999776	.999784	.999792	.999800	.999807	.999815	.999822	.999828	.999835	1	1	2	3	4	4	5	6	7
3.6	.999841	.999847	.999853	.999858	.999864	.999869	.999874	.999879	.999883	.999888	0	1	1	2	2	3	3	4	5
3.7	.999892	.999896	.999900	.999904	.999908	.999912	.999915	.999918	.999922	.999925									
3.8	.999928	.999931	.999933	.999936	.999938	.999941	.999943	.999946	.999948	.999950									
3.9	.999952	.999954	.999956	.999958	.999959	.999961	.999963	.999964	.999966	.999967									

For negative values of z use $\Phi(z) = 1 - \Phi(-z)$